



UNIVERSIDAD DE LAS FUERZAS ARMADAS - ESPE

DATE:

15/06/2026

CLASS:

AWD - 30716

WORKSHOP:

Product Backlog Validation

Nº) NOMBRES Y APELLIDOS..... No/10 Git ☐ Doc ☐  

01) DYLAN ALEXANDER ALVARADO PALACIOS..... 10/10

02) NELSON DAVID ANDINO MARTINEZ..... 10/10

03) JILMAR JHOEL CALDERON CALVA..... 10/10

Features: -6 4/10

Features

Feature	Description	Actor(s)	User Stories	PB Items
F-01	Authentication & Role-Based Access Control — Login, session management, and role-restricted module access.	All Actors	US-01	PB-01, PB-02, PB-03
F-02	Patient Management — Full CRUD of patient records with ID, age, legal representative, and consultation reason validations.	Dentist	US-02, US-03	PB-04, PB-05, PB-06, PB-07, PB-08, PB-17
F-03	Patient Business Logic — Server-side calculations on patient data: pediatric category classification, legal representative check, birthday countdown, consultation time, urgency level, and treatment complexity.	Dentist	US-02, US-03	PB-21, PB-22, PB-24, PB-26, PB-27, PB-31
F-04	Payment Management — Full CRUD of payment records with type (Deposit / Final), method (Cash / Card / Transfer), amount and date validations.	Receptionist	US-04, US-05, US-08	PB-09, PB-10, PB-11, PB-12

F-05	Payment Business Logic — Derived payment status, senior discount eligibility, contact priority score, outstanding patient balance, and daily revenue summary.	Receptionist	US-05, US-08	PB-23, PB-25, PB-28, PB-29
F-06	Inventory Management — Full CRUD of dental supplies with quantity, unit cost, order date, and expiration date validations and automatic status assignment.	Administrator	US-06, US-07	PB-13, PB-14, PB-15, PB-16
F-07	Inventory Analytics & Alerts — Financial analysis (asset valuation, restock budget, expiration losses, capital risk) and critical stock alerts.	Administrator	US-07, US-09	PB-15, PB-30
F-08	Security & Reporting — Encrypted credential storage, statistical dashboard, patient/inventory search, and PDF/Excel export.	All Actors	—	PB-17, PB-18, PB-19, PB-20

7.2.2 Payments — CRUD

Tag	Method	URI	Description
[CRUD]	POST	/fabuladental/payments	Record a new patient payment (Deposit or Final)
[CRUD]	PUT	/fabuladental/payments/:paymentId	Update a recorded payment to correct errors or status
[CRUD]	DELETE	/fabuladental/payments	Delete a payment record by ID (sent in request body)

The GET method is not documented

5.1 Implemented Functional Requirements

- RF-01: Complete CRUD of payments with types (Deposit/Final) and methods (Cash/Transfer/Card) — TypeScript + Prisma
- RF-02: Automatic derivation of payment status (Deposit → Partial, Final → Completed) in Business Logic layer
- RF-03: Complete CRUD of supplies with automatic expiration status calculation — TypeScript + Prisma
- RF-04: Automatic alerts for products with an expiration date ≤ 30 days (NextExpiration status)
- RF-05: Public treatment page accessible without authentication
- RF-06: RESTful APIs for each module (GET, POST, PUT, DELETE) — TypeScript/Express
- RF-07: Business Logic endpoints for financial analysis: asset valuation, restock provisioning, expiration losses, capital risk
- RF-08: Payment history endpoint with calculated status per payment record
- RF-09: Patient payment history endpoint filtered by patient ID
- RF-10: Legal representative validation endpoint for underage patients (PB-21)
- RF-11: Patient Business Logic endpoints — birthday countdown, senior discount, consultation time estimate, contact priority (PB-22–PB-25)
- RF-12: Patient urgency level assessment endpoint (BL Server) (PB-26)
- RF-13: Patient treatment complexity estimation endpoint (BL Server) (PB-27)
- RF-14: Outstanding patient balance summary endpoint per patient (BL Server) (PB-28)
- RF-15: Daily revenue summary endpoint grouped by payment method (BL Server) (PB-29)
- RF-16: Critical stock alert endpoint for inventory management (BL Server) (PB-30)
- RF-17: Pediatric category classification endpoint based on patient date of birth (BL Server) (PB-31)

5.2 Non-Functional Requirements

- RNF-01 Usability: Responsive interface with Bootstrap 5. Compatible with mobile and desktop.
- RNF-02 Performance: CRUD operations do not require full page reload (fetch API). BL communicates with CRUD via HTTP.
- RNF-03 Scalability: PostgreSQL (Supabase) with pgbouncer connection pooling for horizontal scaling.
- RNF-04 Portability: Deployable on GCP or any Node.js-compatible environment.
- RNF-05 Maintainability: Clean separation between CRUD and Business Logic servers. TypeScript strict mode enforced.
- RNF-06 Validation: Backend input validation — 10-digit IDs, positive amounts, non-future dates, valid enums, inventory business rules.
- RNF-07 Type Safety: All backend code compiled with TypeScript strict mode with zero type errors.

URI not implemented: /patients/urgency-level

1.6 Features Overview

treatment

The table below groups all Product Backlog items into the eight high-level features of the system. Each feature is owned by one or more actors and cross-references the corresponding User Stories and PB items.

Feature	Description	Actor(s)	User Stories	PB Items
F-01	Authentication & Role-Based Access Control — Login, session management, and role-restricted module access.	All Actors	US-01	PB-01, PB-02, PB-03
F-02	Patient Management — Full CRUD of patient records with ID, age, legal representative, and consultation reason validations.	Dentist	US-02, US-03	PB-04, PB-05, PB-06, PB-07, PB-08, PB-17
F-03	Patient Business Logic — Server-side calculations on patient data: pediatric category classification, legal representative check, birthday countdown, consultation time, urgency level, and treatment complexity.	Dentist	US-02, US-03	PB-21, PB-22, PB-24, PB-26, PB-27, PB-31
F-04	Payment Management — Full CRUD of payment records with type (Deposit / Final), method (Cash / Card / Transfer), amount and date validations.	Receptionist	US-04, US-05, US-08	PB-09, PB-10, PB-11, PB-12

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URI not implemented: /patients/treatment-complexity

Fabula Dental — Product Backlog v3.0 | Team: CodeExplorers | June 12, 2020

F-05	Payment Business Logic — Derived payment status, senior discount eligibility, contact priority score, outstanding patient balance, and daily revenue summary.	Receptionist	US-05, US-08	PB-23, PB-25, PB-28, PB-29
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URI not implemented: /payments/patients/:id/balance

URI not implemented: /payments/daily-revenue

F-07	Inventory Analytics & Alerts — Financial analysis (asset valuation, restock budget, expiration losses, capital risk) and critical stock alerts.	Administrator	US-07, US-09	PB-15, PB-30
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URI not implemented: /supplies/critical-stock

Feature	Description	Actor(s)	User Stories	PB Items
F-01	Authentication & Role-Based Access Control — Login, session management, and role-restricted module access.	All Actors	US-01	PB-01, PB-02, PB-03
F-02	Patient Management — Full CRUD of patient records with ID, age, legal representative, and consultation reason validations.	Dentist	US-02, US-03	PB-04, PB-05, PB-06, PB-07, PB-08, PB-17
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F-04	Payment Management — Full CRUD of payment records with type (Deposit / Final), method (Cash / Card / Transfer), amount and date validations.	Receptionist	US-04, US-05, US-08	PB-09, PB-10, PB-11, PB-12

F-05	Payment Business Logic — Derived payment status, senior discount eligibility, contact priority score, outstanding patient balance, and daily revenue summary.	Receptionist	US-05, US-08	PB-23, PB-25, PB-28, PB-29
F-06	Inventory Management — Full CRUD of dental supplies with quantity, unit cost, order date, and expiration date validations and automatic status assignment.	Administrator	US-06, US-07	PB-13, PB-14, PB-15, PB-16
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F-08	Security & Reporting — Encrypted credential storage, statistical dashboard, patient/inventory search, and PDF/Excel export.	All Actors	—	PB-17, PB-18, PB-19, PB-20

- 04) ANDRES SEBASTIAN CARDENAS ERAZO..... 10/10
05) KERLY ANDREINA CHUQUI AGUINDA..... 10/10
06) CAMERON VICTORIA DIAZ CAIZALUISA..... 10/10

Validation of team 3:

Features 7/10

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2.2	Specialist Profiles	Desc: Creation of therapist accounts linked to specific departments. AC: Therapists have read/write access to their agenda and read-only cross-visibility to other departments.	5	High
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5.2	Cancellation Penalty Enforcement	Desc: Logic to penalize late cancellations. AC: Canceling within 24 hours triggers a non-refundable 100% penalty warning. The deposit is logged as earned revenue	5	High
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6.2	Downloadable PDF Vouchers	Desc: Generate a digital receipt upon appointment confirmation. AC: PDF must include the Ser salud logo, Date, Time, Amount, Specialist, and a reminder to arrive 15 minutes early.	5	Med
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1.3	Dynamic Services Catalog	Desc: Display the clinical services (Traumatological, Neurological, Respiratory, Sports) directly on the landing page for quick access.	2	Med
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7.1 View All Therapies

GET <http://20.55.45.48:4000/api/terapias>

1. Add a transaction inspection inquiry structural node point path called **Therapies** - **Get All**.
2. Maintain the selection state indicator configuration option parameter fields exactly as standard **GET** context.
3. Supply the endpoint destination reference location address link mapping string: **http://20.55.45.48:4000/api/terapias**.
4. No payload bodies or special token credentials are required. Click **Send** to verify the return array formatting matches collection design guidelines.

7.2 View Therapies by Specialty

GET <http://20.55.45.48:4000/api/terapias?especialidad=Fisioterapia>

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
2.1	Frictionless Patient Login	Desc: Registration asks only for DNI and Full Name to prevent user drop-off. AC: System auto-generates a username (e.g., initial + surname) and sets the DNI as the permanent password.	5	High

6.2 Register User (Patient)

POST <http://20.55.45.48:4000/api/auth/register>

Applies strict validation of the Ecuadorian identification algorithm (Cédula).

2.3	Admin Privilege Management	Desc: Administrator dashboard to manage clinic staff and services. AC: Admin can add new therapy areas and assign specialists dynamically.	3	Med
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10.1 Get All Users

GET <http://20.55.45.48:4000/api/users>

Authorization: Bearer {{token_admin}}

1. Setup a collection tracking testing node and label it **Users - Get All (Admin)**.

3.1	Deferred Medical History	Desc: A non-mandatory clinical history form presented as a gentle prompt before the first booking. AC: Can be skipped by the patient and filled out later by the therapist during the physical session.	8	High
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5.2 Find Many Records (findMany)

POST <http://3.135.164.193:4000/db/query>

Extracts collection records mapped structurally to domain parameters.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
4.1	24-Hour Chrono-logical Validation	Desc: Enforce a strict buffer for all non-emergency appointments. AC: The interactive calendar disables any time slots that are less than 24 hours away from the server's current timestamp.	8	High

9.1 Get Available Time Slots

GET <http://20.55.45.48:4000/api/citas/horarios-disponibles?medicoId=9&fecha=2026-06->
Authorization: Bearer {{token_jwt}}

1. Construct a query monitoring object and label it **Appointments - Get Available Slots**.

4.2	Cross-Department Anti-Overlap	Desc: Prevents a single patient from holding concurrent appointments. AC: If Patient A has Traumatology at 10:00 AM, the system blocks them from booking Neurology at 10:00 AM on the same day.	8	High
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9.1 Get Available Time Slots

GET <http://20.55.45.48:4000/api/citas/horarios-disponibles?medicoId=9&fecha=2026-06->
Authorization: Bearer {{token_jwt}}

1. Construct a query monitoring object and label it **Appointments - Get Available Slots**.

4.3	Internal Patient Referral	Desc: Allows a specialist to schedule a subsequent appointment for their patient. AC: Therapists can book slots in their own calendar or refer the patient directly into an available slot in another department.	5	Med
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9.2 Create Appointment

POST <http://20.55.45.48:4000/api/citas>
Authorization: Bearer {{token_jwt}}

1. Deploy a resource transaction reservation workflow sequence item module and label it **Appointments - Book Appointment**.

07) LENIN ISAAC ERAZO CAMPOS..... 10/10
08) CÉSAR LUIS GALARZA VILLALVA..... 10/10
09) BRAYAN PATRICIO GUALOTUÑA AMAGUAYA... 10/10

Features: (input) **9.0 / 10** (1 issue)

Conflicting URI: In the DELETE /appointments/{appointment_id} section, the example parameter points to a completely different route: /appointment-services/...

DELETE /appointments/{appointment_id}

Deletes an appointment from the database.

Parameters

Parameter	Location	Tipo	Required	Description
appointment_id	/appointment-services/9785f22b-3a89-4945-82a7-77d379756a78	string	Sí	Unique appointment identifier a eliminar

Codes de Respuesta

Code	Description
204	Appointment deleted successfully (no content)
422	Validation error

FEATURE: AUTHENTICATION & USERS	METHOD	URI
User Registration	POST	/auth/register
User Login	POST	/auth/login
Role Management (RBAC)	GET	/sessions/current
Authorization Middleware	GET	/sessions/current

FEATURE: BARBERSHOP MANAGEMENT	METHOD	URI
Create Barbershop	POST	/barbershops

Edit Barbershop	PUT	/barbershops/{shop_id}
View Barbershop Info	GET	/barbershops/{shop_id}
Edit Restriction	PUT	/barbershops/{shop_id}

FEATURE: BARBER MANAGEMENT	METHOD	URI
List Barbers	GET	/barbershops/{shop_id}/members
Manage Barbers	POST	/api/owner/barbers
Assign Barbers to Shop	POST	/barbershops/{shop_id}/members

FEATURE: SERVICES	METHOD	URI
Create Service	POST	/barbershops/{shop_id}/services
Edit Service	PUT	/barbershops/{shop_id}/services/{service_id}
Delete Service	DELETE	/barbershops/{shop_id}/services/{service_id}
List Own Services	GET	/barbershops/{shop_id}/services

Public Service Ownership	GET	/api/customer/services
List Services Restriction	GET	/barbershops/{shop_id}/services

FEATURE: PRODUCTS	METHOD	URI
Create Product	POST	/barbershops/{shop_id}/products
Edit Product	PUT	/barbershops/{shop_id}/products/{product_id}
Delete Product	DELETE	/barbershops/{shop_id}/products/{product_id}
List Products	GET	/barbershops/{shop_id}/products

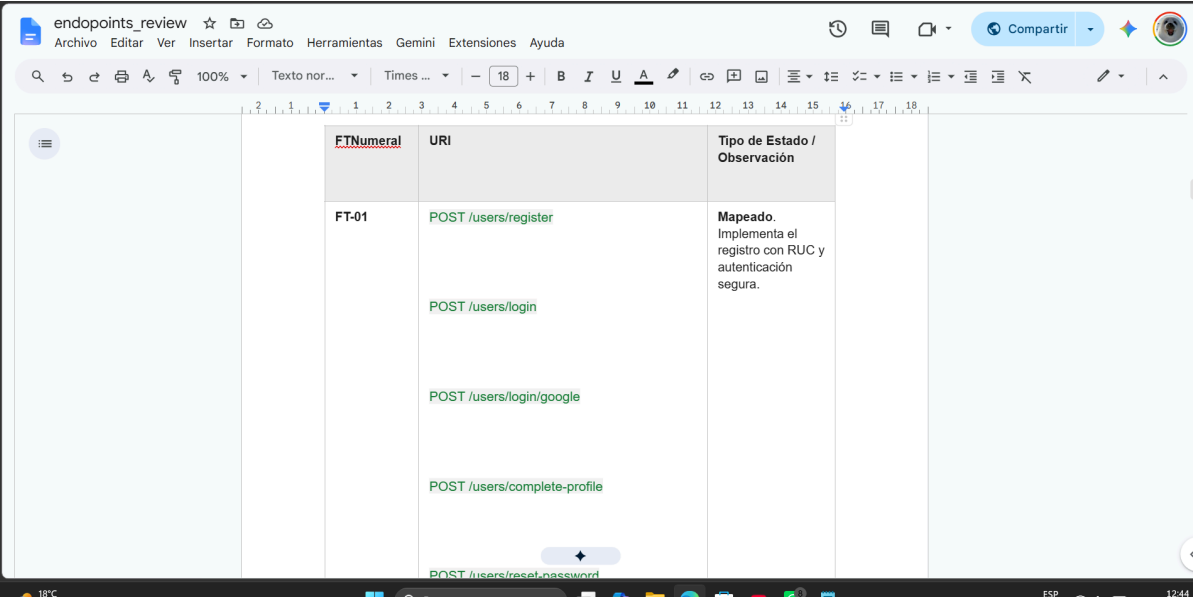
FEATURE: BOOKINGS / APPOINTMENTS	METHOD	URI
Create Appointment	POST	/appointments
View Availability	GET	/api/customer/available-times
Select Service	POST	/appointments/{appointment_id}/services

Barber Calendar	GET	/api/barber/appointments
Owner Dashboard	GET	/api/owner/appointments
Update Appointment Status	PATCH	/api/owner/appointments/{appointment_id}/status
Availability Validation	POST	/appointments

10) GABRIEL ANTHONY MOLINA GALLEGOS..... 10/10

URL:

<https://docs.google.com/document/d/1LnYWdqRkRqRcqdoAG9EvP2BPWPOGm-1lGNNqDdVosKc/edit?usp=sharing>



The screenshot shows a Google Docs interface with a document titled 'endpoints_review'. The document contains a table with three columns: 'FTNumeral', 'URI', and 'Tipo de Estado / Observación'. The table lists several API endpoints, including 'POST /users/register', 'POST /users/login', 'POST /users/login/google', 'POST /users/complete-profile', and 'POST /users/reset-password'. The 'Tipo de Estado / Observación' column contains the text 'Mapeado. Implementa el registro con RUC y autenticación segura.'

FTNumeral	URI	Tipo de Estado / Observación
FT-01	POST /users/register	Mapeado. Implementa el registro con RUC y autenticación segura.
	POST /users/login	
	POST /users/login/google	
	POST /users/complete-profile	
	POST /users/reset-password	

11) RICARDO ANDRES MONGE MIÑO..... 10/10

URL:

<https://docs.google.com/document/d/1LnYWdqRkRqRcqdoAG9EvP2BPWPOGm-1lGNNqDdVosKc/edit?usp=sharing>

FTNumeral	URI	Tipo de Estado / Observación
FT-01	POST /users/register POST /users/login POST /users/login/google POST /users/complete-profile POST /users/reset-password	Mapeado. Implementa el registro con RUC y autenticación segura.

12) ALEJANDRO XAVIER OBANDO ZAPATA..... 10/10

URL:

<https://docs.google.com/document/d/1LnYWdqRkRqRcqdoAG9EvP2BPWPOGm-1lGNNqDdVosKc/edit?usp=sharing>

FTNumeral	URI	Tipo de Estado / Observación
FT-01	POST /users/register POST /users/login POST /users/login/google POST /users/complete-profile POST /users/reset-password	Mapeado. Implementa el registro con RUC y autenticación segura.

13) ESTEBAN ARIEL QUIROGA BENITEZ..... 10/10

14) DAVID JOSUE RODRIGUEZ VILLARROEL..... 10/10

15) ANGEL JEANPIERRE SABANDO MORA..... 10/10

URL:<https://docs.google.com/document/d/1OGXhIXM2BEMKpCJBloOn-quwKto58-69/edit?usp=sharing&ouid=109487110320888441043&rtpof=true&sd=true>

Accounting System – Product Backlog & Features Document

Product Backlog Decomposition into Features by Module

1. Features Summary Table

ID	Feature Name	Primary Module	Sprint(s)	Lead(s)	Effort
FT-01	User Identity & Secure Access	Mod 1	1	D.R / E.Q	7 hrs
FT-02	Workspace & Project Isolation	Mod 1	1	A.S	7 hrs
FT-03	SRI Service Handshake	Mod 2	2-3	E.Q	7 hrs

FT-02 – Workspace & Project Isolation

Field	Detail
Description	Creation of isolated local directories based on specific taxpayer RUCs and periods.
Objective	Prevent cross-contamination between different fiscal periods.

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Module	Module 1 – Configuration & Setup
Associated RFs	RF-02, RF-03
Responsible	Angel Sabando
Business Rules	RH-01 (Tax periods)

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs

3.1 Module 1: Authentication and Base Profile (Taxpayer)

Está viendo cambi

Method	Endpoint / URI	Description / Payload (JSON)
POST	/auth/register	Initial registration on the platform. Body: { email, password, firstName, lastName }
POST	/auth/login	Web app login. Validates against Supabase Auth. Returns: JWT Token + User data.
GET	/taxpayer/profile	Retrieves the authenticated taxpayer's profile information. Header: Authorization: Bearer <token>
PUT	/taxpayer/profile	Updates the taxpayer's profile information. Body: { firstName, lastName, phone }

16) CARLOS ALEXANDER TORRES PINCAY..... 10/10

17) EVELYN HAYDEE VILLARREAL SALGADO..... 10/10

URL:<https://docs.google.com/document/d/1ZnIFeZfH-KQanjoiKeMpUHM7u-TDnITZVTgVnIB1frk/edit?usp=sharing>

Validation Scope

Area	Evidence
Requirements source	@2Requirements/requirements.md
API validation report	@3Documentation/api-validation-report.md
Business rules report	@3Documentation/business-rules-api-test-report.md
Postman collection	postman/American-Latin-Class-API.postman_collection.json
Python analytics collection	postman/American-Latin-Class-Analytics-API.postman_collection.json
JWT/Postman evidence	postman/evidence/postman-local-jwt-auth-evidence.md
Python API evidence	postman/evidence/python-analytics-api-evidence.md

Backlog Item Validations

PBI-01 - Auth, Sessions, RBAC And Private Pages

Field	Validation
Related stories / requirements	US-02, US-07, FR-02, FR-03, FR-04
Acceptance criteria	Google/Auth login creates a secure app session; JWT Bearer access works in Postman; logout revokes the session; protected routes reject anonymous or revoked tokens.
Endpoint evidence	POST /api/v1/auth/login , POST /api/v1/auth/google , GET /api/v1/auth/me , POST /api/v1/auth/logout
Documentation evidence	@3Documentation/oauth-session.md , @3Documentation/postman-token-auth-proof.md
Status	Completed / Validated

PBI-02 - PostgreSQL Schema, Seeders And Repository Abstraction

Field	Validation
Related requirements	FR-05 through FR-14, NFR-03
Acceptance criteria	Normalized PostgreSQL schema exists; seed data loads roles, permissions, branches, dance categories/styles and rules; repository layer isolates persistence.

Definition Of Done Validation

Definition Of Done Item	Evidence	Status
Requirement is documented	@2Requirements/requirements.md	Done
API endpoint or UI workflow exists	@3Documentation/api-documentation.md , @6Code/src , @6Code/frontend , @6Code/python-analytics-api	Done
Business rule is implemented in service layer	@6Code/src/services , @6Code/python-analytics-api/app/services.py	Done

rs/User/Desktop/WEB%203/AWD-30716-Quantum-Code/08PDFDeliverables/build/html/03Documentation/product

a.m. Product Backlog Validation

Field	Validation
Code evidence	@6Code/migrations/001_initial_schema.sql , @6Code/seeders/001_seed.sql , @6Code/prisma/schema.prisma , @6Code/src/repositories
Status	Completed / Validated

a.m. Product Backlog Validation

Definition Of Done Item	Evidence	Status
Data model supports the feature	PostgreSQL migration, Prisma schema, RDS seed records	Done
Automated or manual test evidence exists	Jest, API validation suite, Python unittest, Postman evidence	Done